

ALI KHAN

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EDUCATION

Rutgers Honors College

New Brunswick, NJ

B.S. in Computer Science and Mathematics; GPA: 4.00/4.00

Expected May 2028

Graduate *Economics and Computation (CS 598)*, Design and Analysis of Data Structures and Algorithms I (CS 513)**

Mathematics *Real Analysis I, Mathematical Theory of Probability, Introduction to Stochastic Processes, Stochastic Calculus, Linear Optimization, Combinatorial Theory*, Linear Algebra (proof-based)*, Introduction to Mathematical Reasoning*

Computer Science *Design and Analysis of Computer Algorithms, Introduction to Artificial Intelligence, Data Structures, Introduction to Discrete Structures*

**in progress, Fall 2026*

PUBLICATIONS AND PREPRINTS

A. Khan. “Channels to Targets: A Closed-Form Coordinate System for Silence-Blindness in LLM Observers.” *Workshop on Epistemic Intelligence in Machine Learning (EIML3), NeurIPS 2026*. Code: github.com/akhan531/Control-On-LLMs 2026

RESEARCH EXPERIENCE

Research Fellow, Chancellor’s Signature Initiative SURF

May – Jul 2026

Rutgers–New Brunswick Office for Research | Faculty mentor: Prof. Laurent Burlion

New Brunswick, NJ

- Independently developed a dynamical-systems characterization of multi-agent LLM debate, modeling deliberation as Bayesian belief updating and tracking progress by the divergence between the group’s public belief and the full-information posterior
- Established that a Lyapunov argument certifies convergence under honest disclosure and fails under strategic withholding; validated the model against Llama-3.3-70B debates over 40 seeds
- Demonstrated a misaligned incentive in LLM-judge credit assignment: withholding a pivotal clue until late in the debate earns a higher credit share than early disclosure, and the effect strengthens with judge capability

Undergraduate Researcher

Sep 2025 – May 2026

Mathematical Optimization Research Group | Advisors: Prof. Farzad Yousefian, Fateme Maleki

New Brunswick, NJ

- Designed IncentOSFL, a one-shot variant of an incentivized federated learning framework, by observing that a convex loss drives every client’s local model to the same global optimum, making per-round aggregation redundant
- Reduced per-round communication from full model parameters to data-contribution values alone; validated on MNIST, showing no client drift and convergence of client contributions to a Nash equilibrium independent of initialization
- Built an industrial case study on the UCI Steel Plates Faults dataset demonstrating that the incentivized mechanism outperforms the data-gatekeeper’s selfish optimum for every participant
- Supported by U.S. Department of Energy grants #DE-SC0023303 and #DE-SC0025570

Research Fellow, Aresty Summer Science Program

May – Jul 2025

Swarm Intelligence Lab | PI: Prof. Daniel Burbano Lombana | Mentor: Deze Liu

New Brunswick, NJ

- Analyzed zebrafish obstacle-avoidance behavior under five illumination conditions using visual-input metrics, extracting trajectories with AnimalTA and filtering position data in MATLAB
- Quantified illumination effects on linear speed, opacity, thigmotaxis, and spatial entropy via factorial analysis across single-fish and dyad conditions
- Trained a random forest classifier with recursive feature selection to distinguish obstacle-avoidance behaviors at 90% accuracy

TALKS AND PRESENTATIONS

“Belief Dynamics and Strategic Timing in Multi-Agent LLM Debate.” A. Khan and L. Burlion. Poster and lightning talk, *2026 CSI SURF Research Symposium*, Rutgers–New Brunswick. *Jul 2026*

“IncentOSFL: A Communication-Efficient Variant for Incentivized Federated Learning.” A. Khan, F. Maleki, F. Yousefian. Poster, *2026 ISE Research Day*, Rutgers Department of Industrial and Systems Engineering. *May 2026*

“Convex Regularized Portfolio Optimization.” Directed Reading Program symposium, Rutgers Department of Mathematics. Mentor: Anupam Nayak. *Dec 2025*

“Vision-based Collective Group Formations in Fish Schools.” A. Khan, D. Liu, D. Burbano Lombana. Poster, *2025 Aresty Summer Research Symposium*, Rutgers–New Brunswick. *Jul 2025*

AWARDS, HONORS, AND FELLOWSHIPS

- Chancellor's Signature Initiative Summer Undergraduate Research Fellowship** — Rutgers–New Brunswick Office for Research. Competitive university-wide research fellowship, \$6,000. 2026
- Second Place, Undergraduate Research Poster Presentation** — 2026 ISE Research Day, Rutgers Department of Industrial and Systems Engineering. Awarded for work on incentivized federated learning; \$200 prize. 2026
- Aresty Summer Science Program** — Aresty Research Center for Undergraduates, Rutgers University. Competitive summer research fellowship, \$3,000. 2025
- Dean's Scholarship** — Rutgers Honors College. Merit scholarship, \$5,000 per semester. 2024 – present
- Dean's List** — Rutgers School of Arts and Sciences. All semesters. 2024 – present

TEACHING EXPERIENCE

- Part-Time Lecturer** Sep 2025 – Present
Department of Computer Science, Rutgers University New Brunswick, NJ
- Lead a weekly recitation section of approximately 50 students and hold weekly office hours
- CS 205, Introduction to Discrete Structures I* — Prof. Akash Sengupta Fall 2026, Spring 2026
- CS 205, Introduction to Discrete Structures I* — Prof. Jingjin Yu Fall 2025
- Grader** Jun 2025 – Present
Department of Mathematics, Rutgers University New Brunswick, NJ
- Math 300, Introduction to Mathematical Reasoning* — Prof. Juan Pablo Mejía-Ramos Fall 2026
- Math 300, Introduction to Mathematical Reasoning* — Prof. Christopher Alexander Fall 2026
- Math 478, Introduction to Stochastic Processes* — Prof. Heeyoung Kwon Spring 2026
- Math 477, Mathematical Theory of Probability* — Prof. Joshua Salamon Fall 2025
- Math 477, Mathematical Theory of Probability* — Prof. Triet Pham Summer 2025

SELECTED PROJECTS

- Equilibrium in a Universal Basic Income Economy** | *Math 490, Prof. Kim Weston* May 2026
- Modeled a continuous-time market with heterogeneous CARA agents receiving stochastic income transfers, deriving the state-price deflator from martingale conditions on discounted assets
 - Solved the agents' dynamic optimization via an HJB equation under an exponential ansatz, obtaining closed-form optimal portfolio and initial consumption
 - Imposed stock and initial-consumption market clearing to reduce equilibrium characterization to a PDE system for the agents' value functions
- Regime-Robust Signal Weighting** | *CS 494 Independent Study, Prof. Amélie Marian* Spring 2026
- Adapted the Disparity Compensation Algorithm from explainable fair ranking to portfolio construction by reframing demographic disparity as a gap in risk-adjusted performance across volatility regimes
 - Converted the mechanism from additive selection bonuses to multiplicative signal weights; implemented an end-to-end backtest over 2010–2024 DJIA data with rolling-median regime detection and dollar-neutral portfolio construction
 - Code: github.com/akhan531/CS494-Project
- Genomic Variant Triage with Evo2** | *Python, Next.js, TypeScript, Modal, Anthropic API* Feb–Mar 2026
- Built a variant-interpretation tool scoring single nucleotide variants by delta log-likelihood from the Evo2 7B genome foundation model over 8,192 bp UCSC reference windows, served on Modal H100 GPUs
 - Designed a splicing-disruption module binning canonical splice-site risk by exon–intron boundary distance from the Ensembl REST API, capturing pathogenicity the raw likelihood score misses
 - Developed a VUS triage pipeline ranking panels by a weighted composite of pathogenicity, splice severity, cancer-gene tier, and confidence, with a 29-gene annotation database, live ClinVar context, sentence-transformer RAG, and LLM-generated tumor board notes; covered by 35 unit tests
 - Code: github.com/akhan531/Biotech-AI-Tool
- Finite State Automata Simulator** | *JavaScript, TypeScript, Vis.js* Jul 2025
- Built an interactive deterministic automata simulator with real-time state visualization; deployed to the Rutgers CS Department as an official instructional tool supporting 200+ students

TECHNICAL SKILLS

Languages: Python, MATLAB, C++, Java, JavaScript, SQL
Scientific Computing: NumPy, Pandas, Scikit-Learn, CVXPY, Matplotlib | **Tools:** Git, L^AT_EX, Jupyter, Mathematica